Topic 5 Differential Calculus

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5.1 Differentiation and Stationary Points

5.1 Differentiation and Stationary Points

Definition 5.1

Let $f : I \longrightarrow \mathbb{R}$, where I is an open interval. The <u>derivative</u> of f at $a \in I$ is the slope of the tangent line to the graph of f at $(a, f(a))$.

This value is given by the limit:

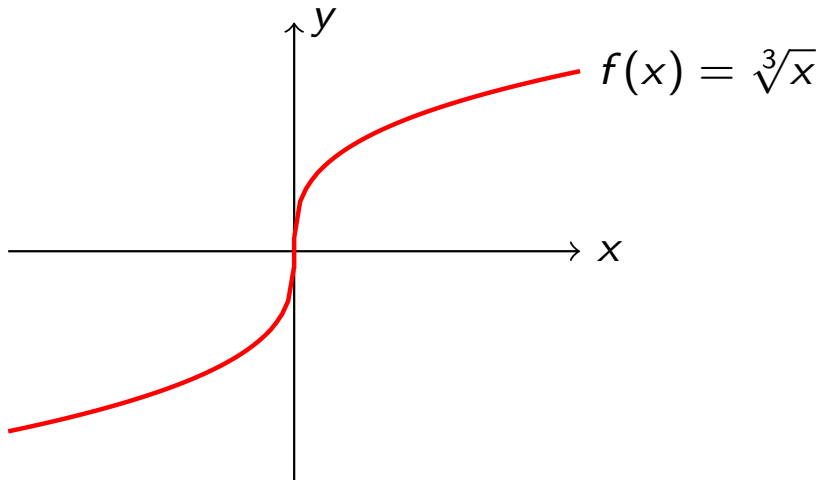
$$f'(a) = \lim_{h \longrightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

When this limit *exists*, f is said to be <u>differentiable</u> at a.

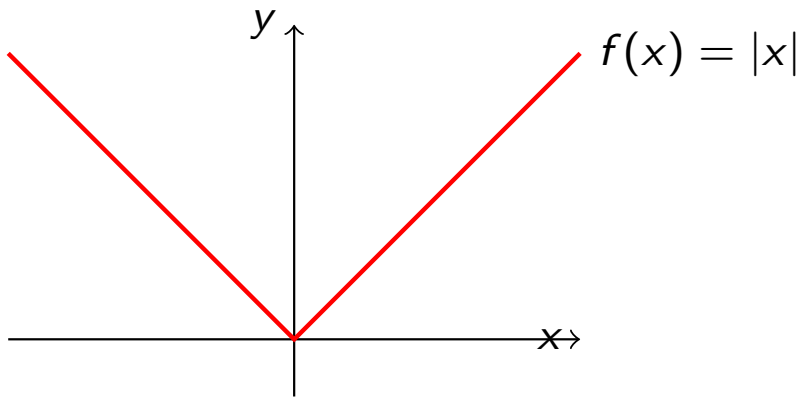
5.1 Differentiation and Stationary Points: Continuity and Differentiability

- There are two typical reasons why *nice* functions fails be differentiable.

1. The tangent line is vertical.



2. The graph has a "sharp" point.



5.1 Differentiation and Stationary Points: Derivatives we "know"

- Another notation for $f'(x)$ is $\frac{d}{dx}[f(x)]$. This is particularly useful when $f(x)$ does not have a well known "name".

$$\frac{d}{dx}[x^a] = ax^{a-1} \qquad \text{for every } a \in \mathbb{R} \setminus \{0\}$$

$$\frac{d}{dx}[\sin(x)] = \cos(x) \qquad (\text{or } \sin'(x) = \cos(x))$$

$$\frac{d}{dx}[\cos(x)] = -\sin(x) \qquad (\text{or } \cos'(x) = -\sin(x))$$

$$\frac{d}{dx}[e^x] = e^x \qquad (\text{or } \exp'(x) = \exp(x))$$

$$\frac{d}{dx}[\log(x)] = \frac{1}{x} \qquad (\text{or } \log'(x) = \frac{1}{x})$$

5.1 Differentiation and Stationary Points

Example 5.2

Let $f : \mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}$ with $f(x) = 2e^x + 3x^{-7}$. Find $f'(x)$.

5.1 Differentiation and Stationary Points: Linearity

Theorem 5.3

Let f and g be differentiable at x and let $c \in \mathbb{R}$.

- $\dfrac{d}{dx}[cf(x)] = c\dfrac{d}{dx}[f(x)]$
- $\dfrac{d}{dx}[f(x) + g(x)] = \dfrac{d}{dx}[f(x)] + \dfrac{d}{dx}[g(x)]$.

5.1 Differentiation and Stationary Points: Linearity

Homework 40

For $f : [0, \infty) \longrightarrow \mathbb{R}$ *with* $f(x) = -9\cos(x) + 2\sqrt{x}$, *find* $f'(x)$.

5.1 Differentiation and Stationary Points

- In addition to using our known derivatives, we also have some techniques to deal with more complicated functions

Theorem 5.4 (Product rule)

If f and g are differentiable at x, then so is the product fg, and:

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$$

Theorem 5.5 (Quotient rule)

If f and g are differentiable at x and if $g(x) \neq 0$, then $\frac{f}{g}$ is differentiable at x and:

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}.$$

Example 5.6

For $f\colon (0,\infty) \longrightarrow \mathbb{R}$ with $f(x) = (x^4 - 3x)\log(x)$, find $f'(x)$.

Homework 41

For $f : \mathbb{R} \longrightarrow \mathbb{R}$ *with* $f(x) = e^x(2x^3 - x^2)$, *find* $f'(x)$.

5.1 Differentiation and Stationary Points

Example 5.7

If $y = \dfrac{x^3}{x^2+1}$ for all $x \in \mathbb{R}$, find $\dfrac{dy}{dx}$.

5.1 Differentiation and Stationary Points

Homework 42

If $f : \mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}$ *with* $f(x) = \dfrac{e^x}{x^3}$, *find* $f'(x)$.

5.1 Differentiation and Stationary Points: Derivatives of Trig Functions

- Using our known derivatives and the product/quotient rule we can find derivatives of some of our trigonometric functions:

Example 5.8

Find $\tan'(x)$.

Example continued

5.1 Differentiation and Stationary Points: Derivatives of Trig Functions

Homework 43

Find $\sec'(x)$.

5.1 Differentiation and Stationary Points: Chain Rule

Example 5.9

Find $\dfrac{dy}{dx}$ if $y = \sin(3x^2 + 8)$.

5.1 Differentiation and Stationary Points: Chain Rule

Theorem 5.10 (The Chain Rule)

If g is differentiable at x and f is differentiable at $g(x)$, then the composition $f \circ g$ is differentiable at x and:

$$\frac{d}{dx}\left[f(g(x))\right] = f'(g(x))\, g'(x).$$

5.1 Differentiation and Stationary Points: Chain Rule

5.1 Differentiation and Stationary Points: Chain Rule

Homework 44

Find $\dfrac{dy}{dx}$ *if* $y = \sqrt{e^x + 4}$.

5.1 Differentiation and Stationary Points

Homework 45

Use the chain rule and the product rule to prove the quotient rule.

Hint: *Write* $\dfrac{f(x)}{g(x)}$ *as* $f(x)h(g(x))$ *where* $h(x) = \frac{1}{x}$.

5.1 Differentiation and Stationary Points: Sketching

- What information can we learn from a function about how it *looks* when we sketch it?

5.1 Differentiation and Stationary Points

Definition 5.11

A function f defined on an interval I is:

- <u>increasing</u> on I if when $a < b$ then $f(a) < f(b)$.
- <u>decreasing</u> on I if when $a < b$ then $f(a) > f(b)$.

- Where is the function $f(x) = \frac{x^3}{x^2+1}$ increasing/decreasing on $\mathbb{R}$?

5.1 Differentiation and Stationary Points

5.1 Differentiation and Stationary Points

Theorem 5.12

Let f be differentiable on the interval $I = [a, b]$.

1. *If $f'(x) > 0$ for all $x \in (a, b)$ then f is increasing on I.*
2. *If $f'(x) < 0$ for all $x \in (a, b)$ then f is decreasing on I.*

5.1 Differentiation and Stationary Points

- Question: What does it mean when $f'(x) = 0$?

5.1 Differentiation and Stationary Points

Definition 5.13

A <u>stationary point</u> of f is any point $(x, f(x))$ such that $f'(x) = 0$.

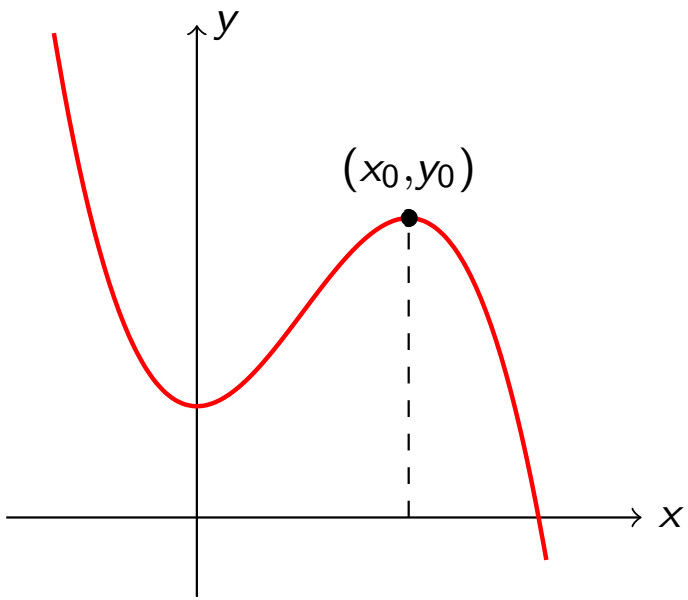
5.1 Differentiation and Stationary Points

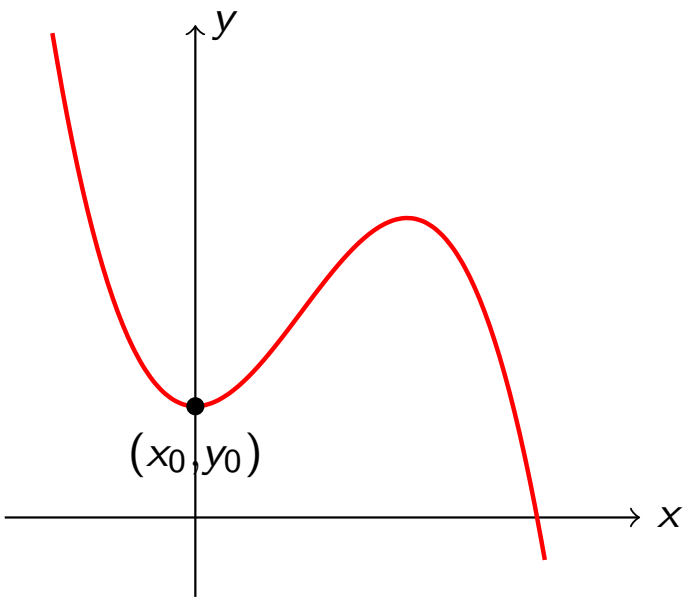
- Why should we care about *stationary* points?

5.1 Differentiation and Stationary Points: Local Extrema and Stationary Points

Definition 5.14

- A function f is said to have:
  - a <u>local maximum</u> at (x_0, y_0) when there is an open interval J containing x_0 on which $y_0 = f(x_0)$ is the largest value.
  - a <u>local minimum</u> at (x_0, y_0) when there is an open interval J containing x_0 on which $y_0 = f(x_0)$ is the smallest value.

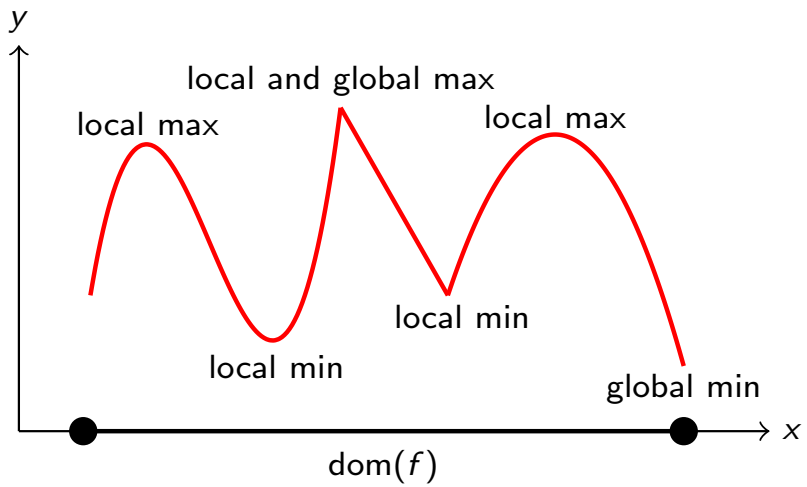




5.1 Differentiation and Stationary Points: Local Extrema and Stationary Points

- Points where a function attains its overall largest and smallest values are called its global extrema.
- The diagram illustrates two types of local extrema:

  1. Points where the graph has a stationary point
  2. Points where the function is not differentiable (has a sharp point).

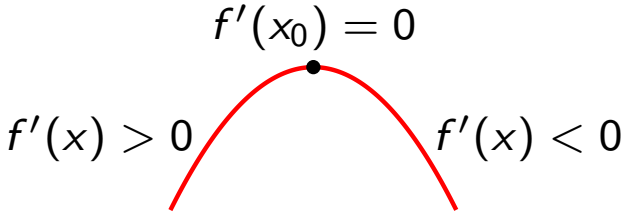


- We focus on the first type:

5.1 Differentiation and Stationary Points: Types of Stationary Points

- A stationary point can be:

1. a local maximum:



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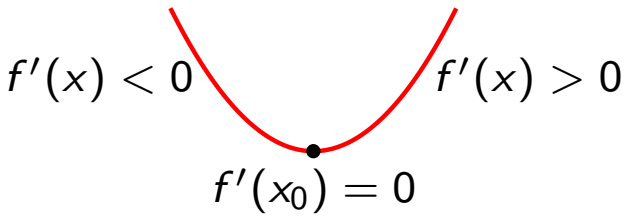
2. a local minimum:



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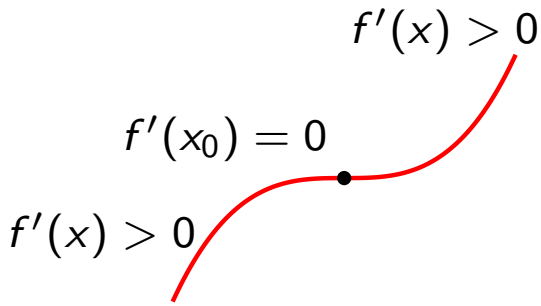
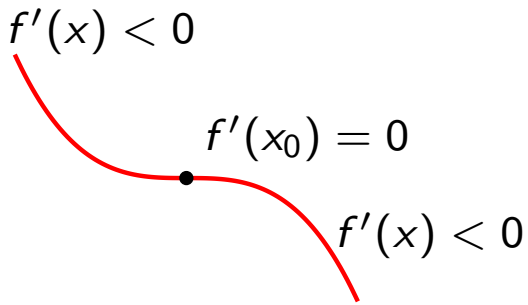
3. or neither:





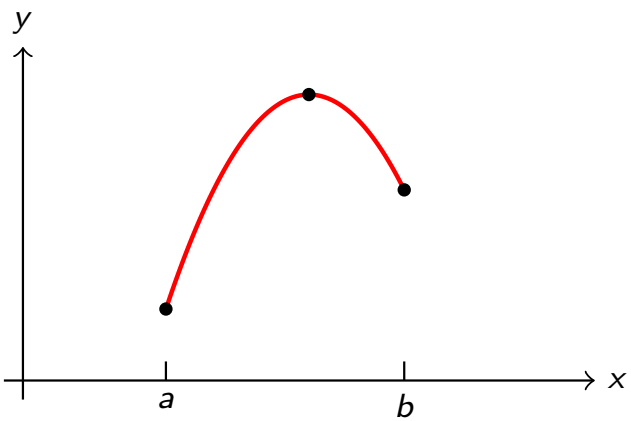
5.1 Differentiation and Stationary Points: Stationary Points

Homework 46

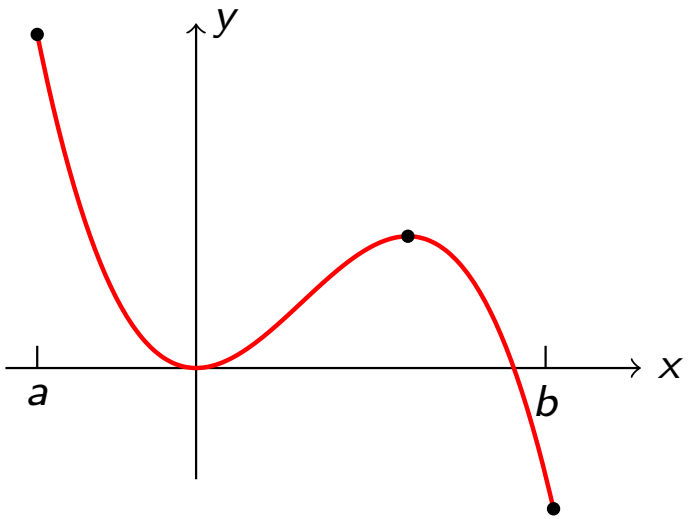
Find the stationary points of the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ *with* $f(x) = e^{-(x-1)^2}$. *On what intervals is* f *increasing? Decreasing?*

5.1 Differentiation and Stationary Points: Stationary Points

- To find the global maximum or minimum of a function f defined on an interval $[a, b]$, it is important to consider:
  - the value of f at any stationary points of f and
  - the value of f at the endpoints of the domain $[a, b]$



Global max at a stationary point $(a_0, f(a_0))$, where $a_0 \in (a, b)$.



Global max is at endpoint $(a, f(a))$, not a stationary point.

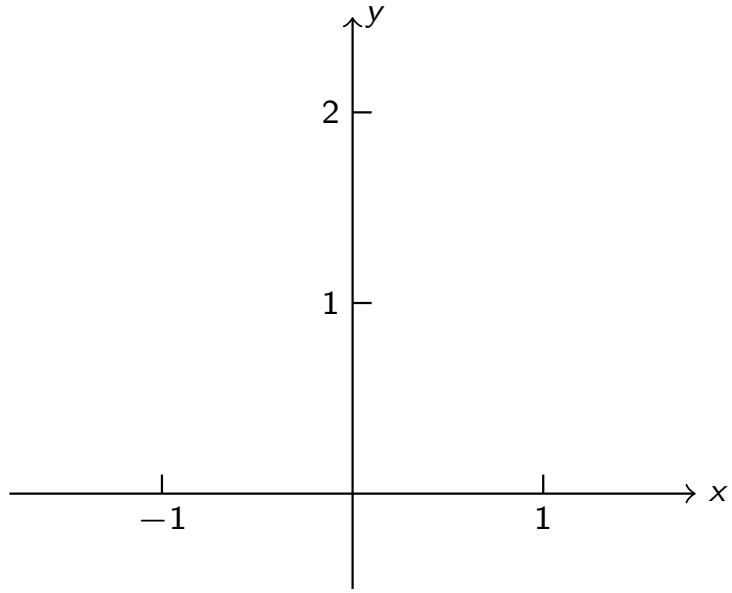
5.1 Differentiation and Stationary Points: Extrema

Example 5.15

Find the global maximum and minimum of

$$f(x) = \sqrt{(x^2+1)(2-x^2)}$$

on its implied domain.



Example continued

5.2 Second and Higher Order Derivatives

5.2 Second and Higher Order Derivatives

- If f' is differentiable at x, differentiating again gives the <u>second derivative</u> of $f(x)$ at x, which is denoted $f''(x)$.
- We say that f is <u>twice differentiable</u> at x.
- Several different notations are commonly used for the second derivative:

Definition 5.16

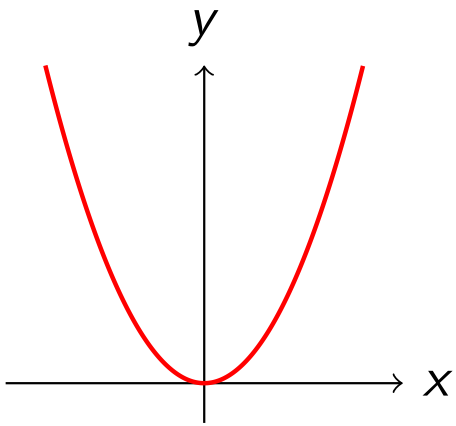
$$f''(x) = \frac{d}{dx}\left[f'(x)\right] = \frac{d}{dx}\left(\frac{d}{dx}f(x)\right) = \frac{d^2}{dx^2}\left[f(x)\right].$$

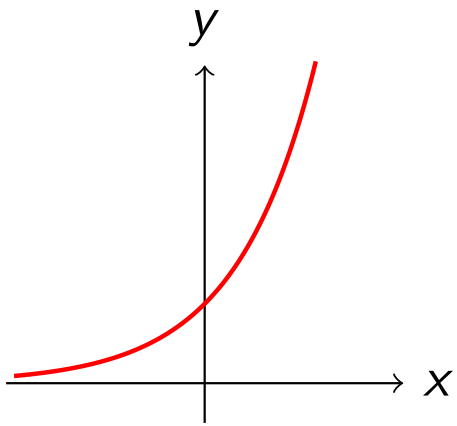
or putting $y = f(x)$:

$$\frac{d}{dx}\left[\frac{dy}{dx}\right] = \frac{d^2y}{dx^2}.$$

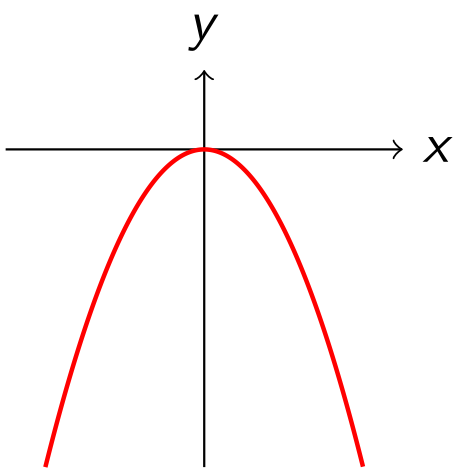
5.2 Second and Higher Order Derivatives: Concavity

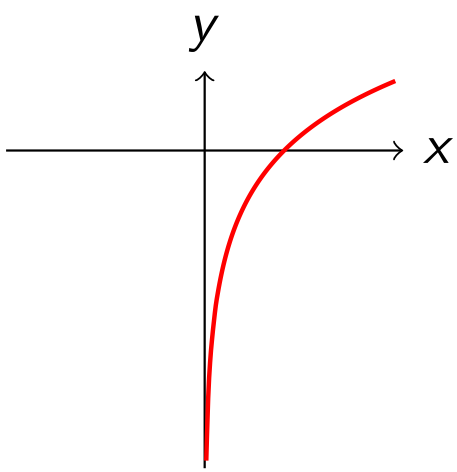
- Concave up and concave down are geometric terms that describe the direction of curvature of the graph of a function.





Graphs of x^2 and e^x are concave up.





Graphs of $-x^2$ and $\log(x)$ are concave down.

5.2 Second and Higher Order Derivatives: Concavity

- For twice differentiable functions, we can determine concavity using the second derivative:

Theorem 5.17

Suppose f is twice differentiable on an interval I.

1. *If $f''(x) > 0$ for all x in I, then f is concave up on I.*
2. *If $f''(x) < 0$ for all x in I, then f is concave down on I.*

5.2 Second and Higher Order Derivatives: Concavity

Example 5.18

Use the second derivative to determine where the following functions $f, g, h : \mathbb{R} \longrightarrow \mathbb{R}$ are concave up and where they are concave down.

(a) $f(x) = x^4$. (b) $g(x) = x^3$. (c) $h(x) = x^{\frac{1}{3}}$.

Example continued

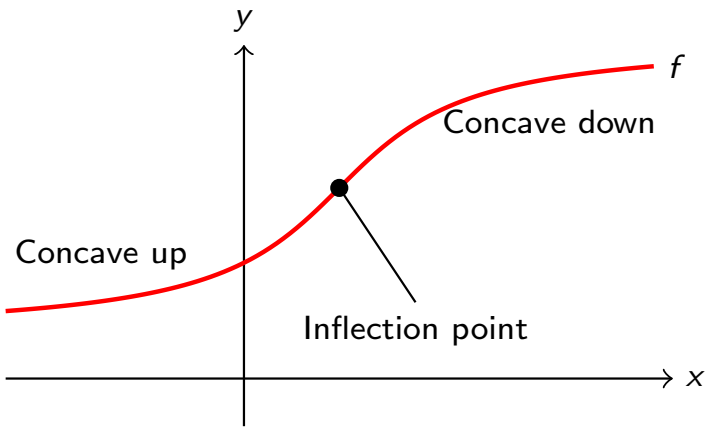
5.2 Second and Higher Order Derivatives: Inflection Points

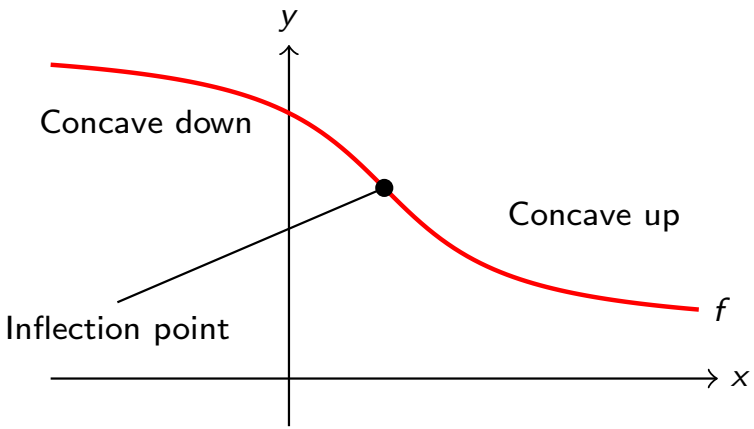
5.2 Second and Higher Order Derivatives: Inflection Points

Definition 5.19

A <u>point of inflection</u> is a point $(x_0, f(x_0))$ on the graph of f where f changes between being concave up and concave down.

Note that $x_0 \in \mathrm{dom}(f)$.



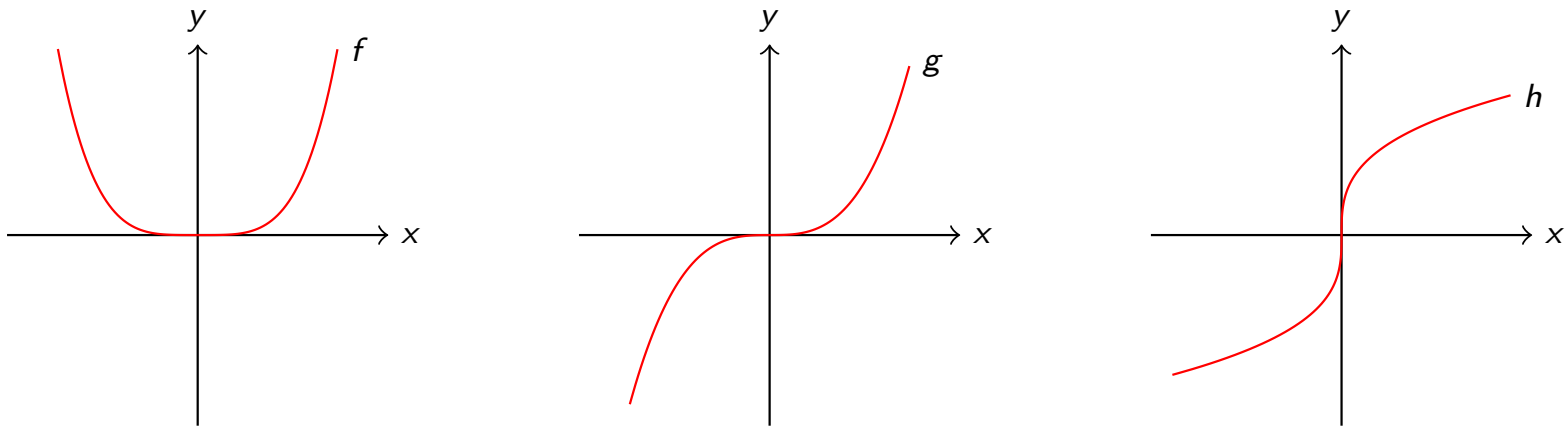


5.2 Second and Higher Order Derivatives: Inflection Points

Example 5.20

Use Example 5.18 to find any points of inflection of $f, g, h : \mathbb{R} \longrightarrow \mathbb{R}$ given by:

(a) $f(x) = x^4$. (b) $g(x) = x^3$. (c) $h(x) = x^{\frac{1}{3}}$.



5.2 Second and Higher Order Derivatives: Inflection Points

- It is NOT enough to simply solve $f''(x) = 0$ to find points of inflection, because:
  - There are cases where $f''(x) = 0$ but x is not a point of inflection (see Example 5.20(a)).
  - There are cases where concavity changes at a point missing from the domain of f'' (see Example 5.20(c)).
- There are also cases where concavity changes at a point missing from the domain of f (see Example 5.24).

5.3 Graph Sketching

5.3 Graph Sketching

Asymptotes are straight lines the graph of a function approaches in some important way. For graph sketching, 2 types are of interest:

- Vertical asymptotes are vertical lines with equation $x = \alpha$ such that the graph approaches either ∞ or $-\infty$ as x approaches α from at least one side of α (possibly both).
- Oblique asymptotes are sloping lines with equation $y = \alpha x + \beta$ that the graph approaches eitheras x approaches $\pm\infty$ When $\alpha = 0$ we get a horizontal asymptotes

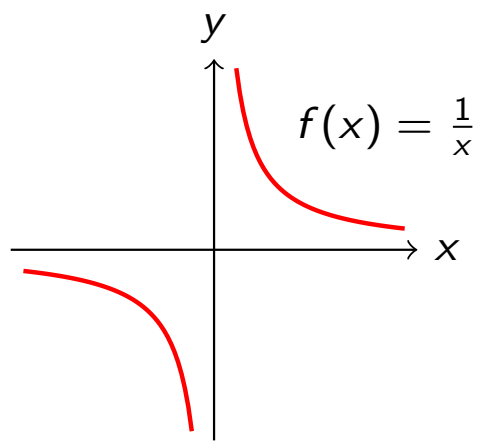
Combining knowledge of asymptotes with knowledge of extrema and concavity helps us to understand the graphs of functions.

5.3 Asymptotes and Graph Sketching: Vertical Asymptotes

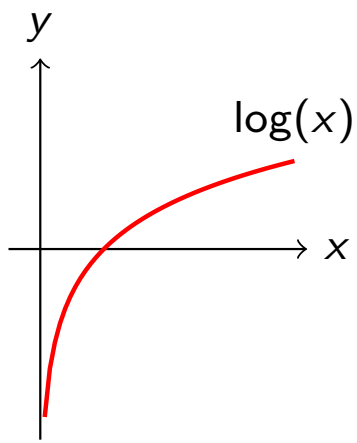
- We often encounter vertical asymptotes when a 'zero denominator' occurs. That is, when

$$f(x) = \frac{g(x)}{h(x)}$$

and $h(x) = 0$ for some values of x.



- As we will see in Example 5.21, a 'zero denominator' does not guarantee a vertical asymptote.
- Vertical asymptotes may also arise in situations that do not involve quotients.
- **Example:** The function $f(x) = \log(x)$ does not involve quotients, but it has a vertical asymptote at $x = 0$.

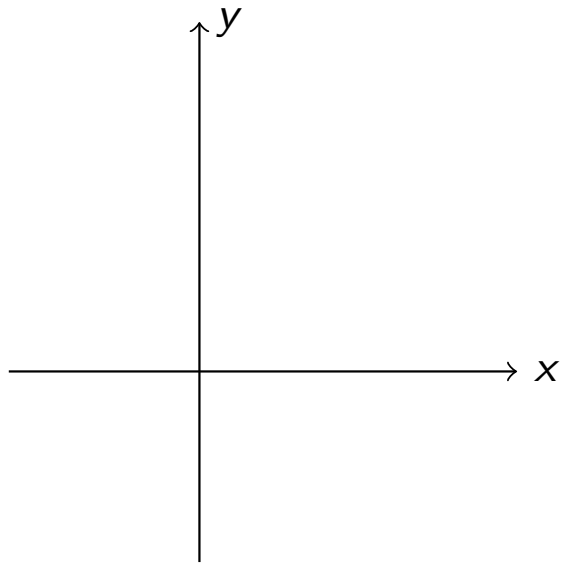


5.3 Asymptotes and Graph Sketching: Vertical Asymptotes

Example 5.21

Find the vertical asymptotes of $f : \mathbb{R} \setminus \{-1, 1\} \longrightarrow \mathbb{R}$ defined by:

$$f(x) = \frac{x^2 + 2x + 1}{x^2 - 1}$$



5.3 Asymptotes and Graph Sketching: Oblique Asymptotes

5.3 Asymptotes and Graph Sketching: Oblique Asymptotes

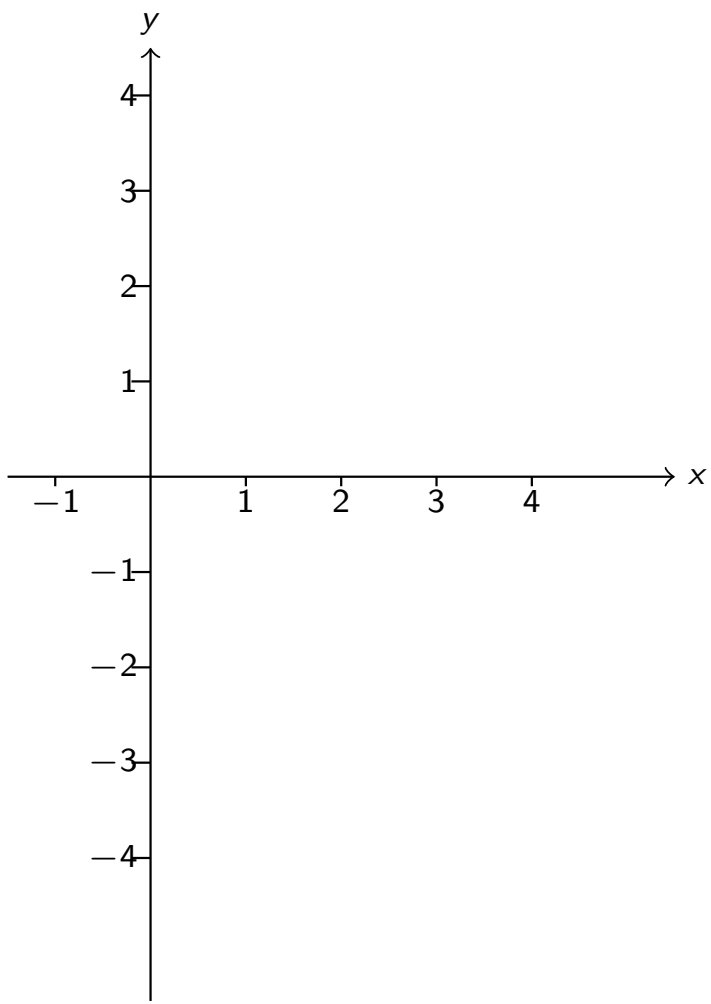
- Sometimes the graph of a function approaches an <u>oblique asymptote</u> $y = \alpha x + \beta$ either as x approaches $\pm\infty$.
- These asymptotes often occur for rational functions – functions of the form

$$f(x) = \frac{p(x)}{q(x)}$$

  where p and q are polynomials.
- To find an oblique asymptote of a rational function, we apply polynomial long division.
- Polynomial long division also makes it easier to analyse the behaviour of the function as $x \longrightarrow \pm\infty$ and helps us find stationary and inflection points.

Example 5.22

Find and sketch any asymptotes of $f(x) = \dfrac{2x^3 - 3x^2 + 2x - 2}{x^2 + 1}$.

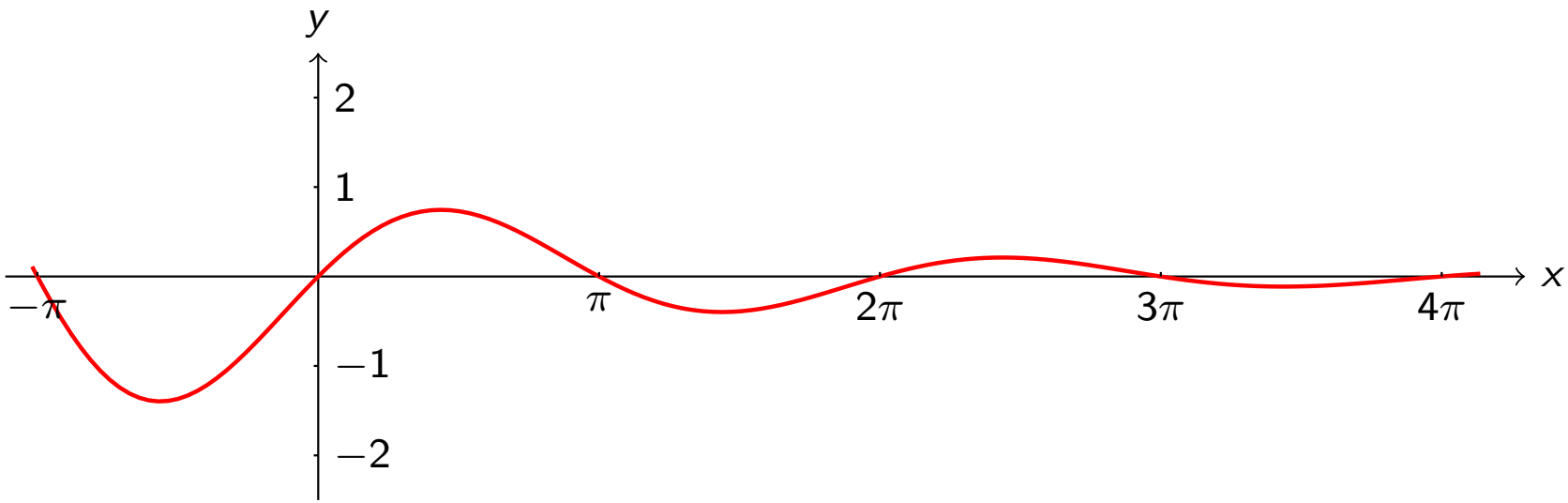


5.3 Asymptotes and Graph Sketching: Oblique Asymptotes

- It is incorrect to think of an asymptote as "a line that may not be crossed".
- This is true for vertical asymptotes.
- A function may cross a horizontal or oblique asymptote many times.

Example 5.23

What is the horizontal asymptote of $f(x) = e^{-\frac{1}{5}x}\sin(x)$?



5.3 Asymptotes and Graph Sketching

Homework 47

Find and sketch any asymptotes of $f(x) = \dfrac{2x^2 + 3x - 2}{x^2 - 1}$.

5.3 Asymptotes and Graph Sketching: Graph Sketching

Example 5.24

Consider the function $f(x) = \dfrac{x^2}{x+1}$. Find:

1. the domain of f;
2. the asymptotes of f;
3. the x and y intercepts of the graph $y = f(x)$;
4. the stationary points of f;
5. the intervals where f is increasing;
6. the intervals where f is decreasing;
7. the intervals where f is concave up;
8. the intervals where f is concave down;
9. the local maxima/minima of f;
10. the points of inflection of f.

Now use all of this information to sketch the graph $y = f(x)$.

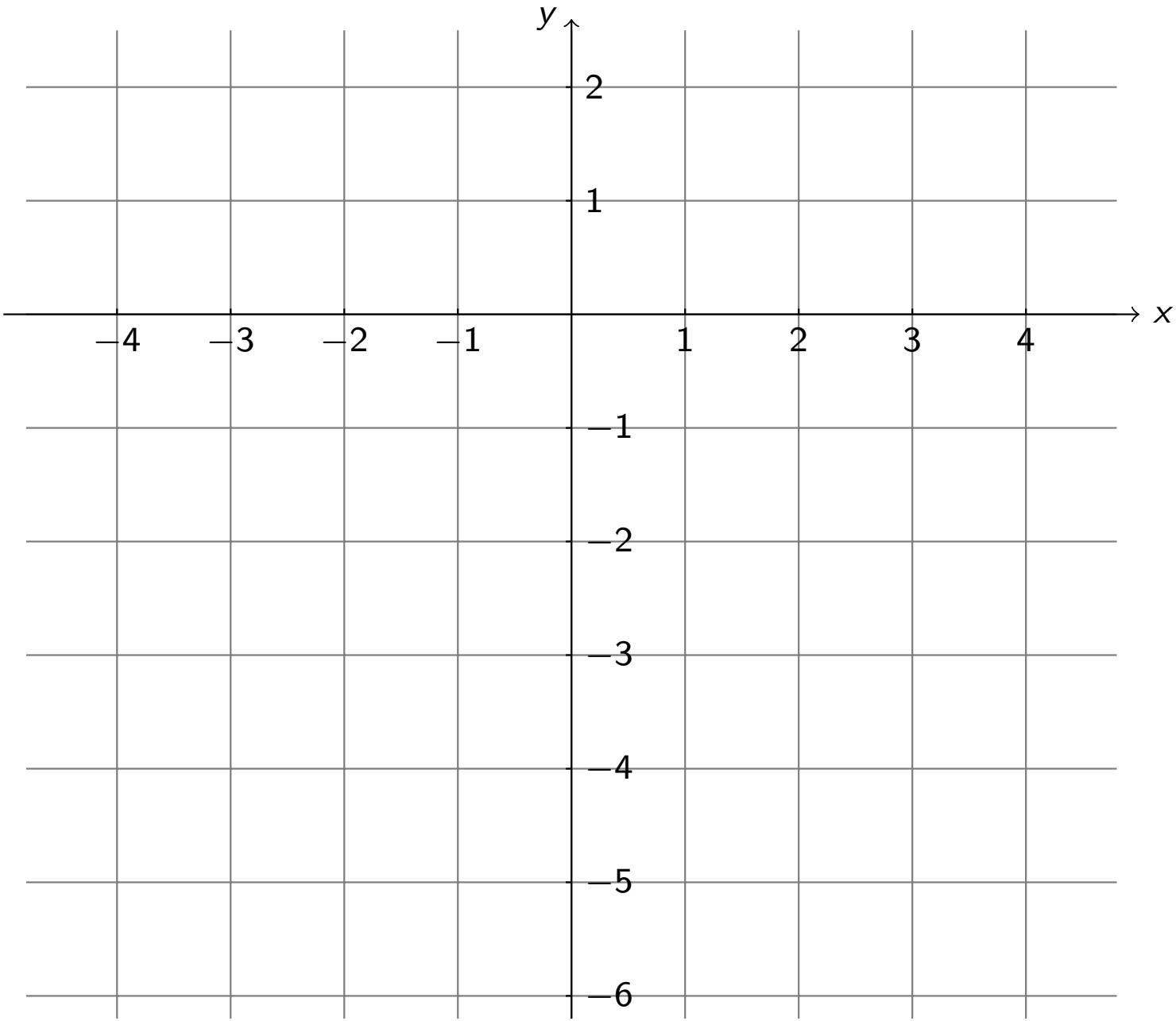
... this looks like a great exam question.

5.3 Asymptotes and Graph Sketching: Graph Sketching

Example continued

Example continued

5.3 Asymptotes and Graph Sketching: Graph Sketching



5.4 Implicit Differentiation

5.4 Implicit Differentiation

Example 5.25

Find a formula for $\dfrac{dy}{dx}$ where

$$x^2 y = 1.$$

5.4 Implicit Differentiation

Example 5.26

Assuming y is a function of x, find $\dfrac{d}{dx}\left[xe^y\right]$

5.4 Implicit Differentiation

- Until now, when finding the derivative $\frac{dy}{dx}$ we knew (or could express) y as a function of of x and apply differentiation rules.
- However, sometimes we cannot express y as a function of x.

Example 5.27

For the curve $C = \{(x, y) \in \mathbb{R}^2 \mid x^4 + y^2 = 1 + x^2 y\}$, can you express y as a function of x?

- This Cartesian equation describes an implicit dependence of y on x.
- Despite this, we can still find the derivative $\frac{dy}{dx}$ using a technique called implicit differentiation.

5.4 Implicit Differentiation

For a Cartesian equation involving both x and y where y depends implicitly on x, the steps are as follows:

1. Differentiate each side with respect to x.
2. Use the usual differentiation rules to simplify each side.
3. Rearrange to solve for $\dfrac{dy}{dx}$. This will usually give a formula for $\dfrac{dy}{dx}$ in terms of both x and y.

Note: When differentiating y, we must treat it as a function of x, so:

- we must apply the chain rule to terms of the form $g(y)$.
- we must apply the product and chain rules rule to terms of the form $f(x)g(y)$.
- etc ...

5.4 Implicit Differentiation

Example 5.28

Assuming y is a function of x, find the following derivatives:

1. $\dfrac{d}{dx}\left[\sin(y)\right]$
2. $\dfrac{d}{dx}\left[y^3\right]$
3. $\dfrac{d}{dx}\left[xe^y\right]$
4. $\dfrac{d}{dx}\left[x^2\log(y)\right]$

Example continued

5.4 Implicit Differentiation

- For our original problem, we cannot solve for y in terms of x, so we use implicit differentiation to find $\dfrac{dy}{dx}$.

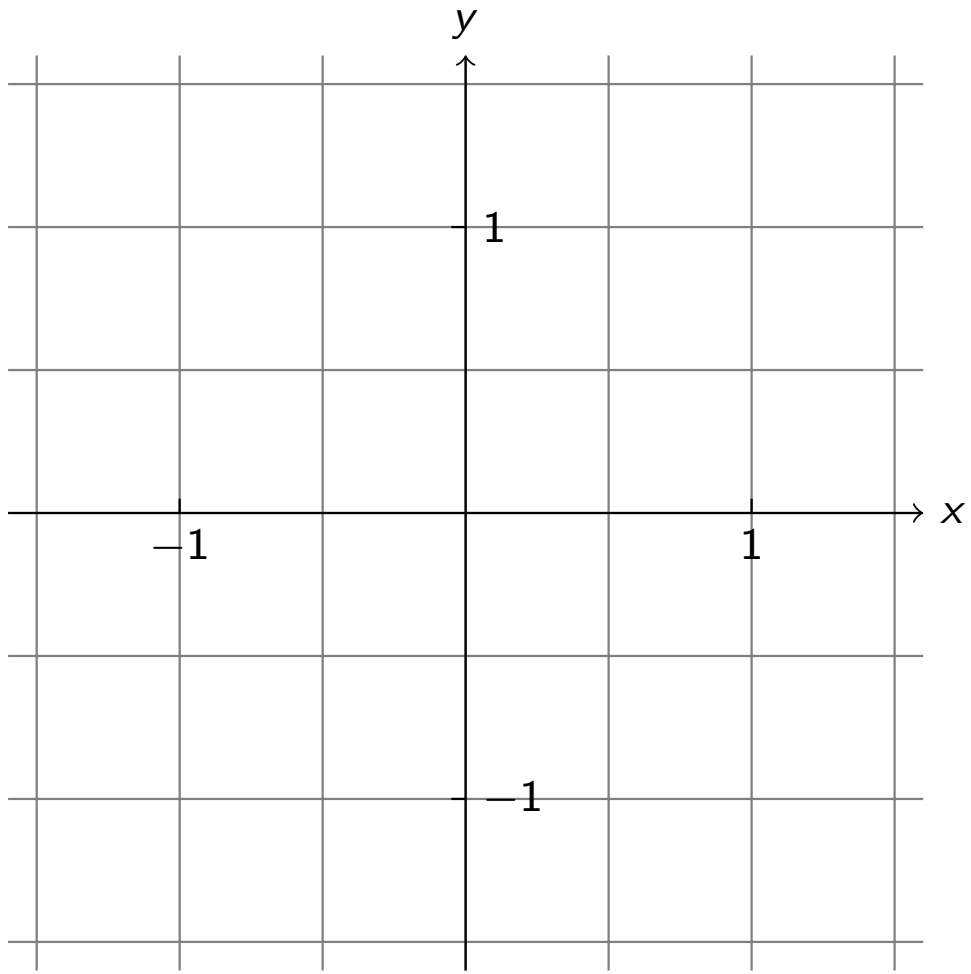
Example 5.29

For the curve $C = \{(x, y) \in \mathbb{R}^2 \mid x^4 + y^2 = 1 + x^2 y\}$:

1. Find $\dfrac{dy}{dx}$.
2. Find the x and y intercepts and hence find the gradient of C at each intercept.
3. Check that $(-1, 1), (1, 1) \in C$ and hence find the gradient of C at each of these points.
4. Use this information to draw a rough sketch of C.

5.4 Implicit Differentiation

5.4 Implicit Differentiation



5.4 Implicit Differentiation: Vertical and Horizontal Tangents

- Implicit differentiation typically gives a quotient: $\dfrac{dy}{dx} = \dfrac{u}{v}$.

Theorem 5.30

Suppose implicit differentiation gives

$$\frac{dy}{dx} = \frac{u}{v}$$

on a curve $C \subseteq \mathbb{R}^2$. Then

1. *At points where $u = 0$ and $v \neq 0$, the tangent line is horizontal.*
2. *At points where $v = 0$ and $u \neq 0$, the derivative is not defined because the tangent line is vertical.*

- Note at points where $u = v = 0$, further analysis is required. Tangent could be vertical, horizontal, sloping or undefined!

5.4 Implicit Differentiation: Vertical and Horizontal Tangents

Example 5.31

For the curve in $C = \{(x, y) \in \mathbb{R}^2 \mid x^3 + y^2 = x^2\}$, find any points at which we can be sure the tangent line is:

1. Vertical
2. Horizontal

Example continued

Example continued

5.4 Implicit Differentiation: Derivatives of Inverse Functions

We can use implicit differentiation to find the derivatives of inverses of well known functions. For mutual inverses like exp and log we already know the derivative of both functions. In other cases, like the trigonometric functions, for example, we need to use the inverse relationship to discover the derivative of the inverse.

In this section, we use implicit differentiation to find the derivatives of the inverses of the standard trigonometric functions.

5.4 Implicit Differentiation: Derivatives of Inverse Functions

- To find the derivative of $y = \log(x)$, we could rewrite this relation as

$$e^y = x$$

  and apply implicit differentiation:

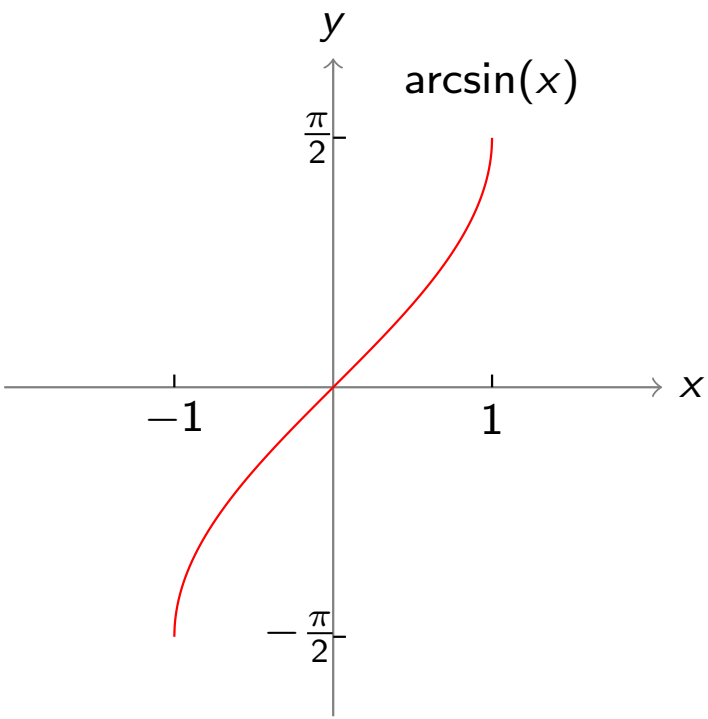
- This gives the well known derivative of log.
- Note that we needed to use the relation $y = \log(x)$ to express our answer as a function of x.

5.4 Implicit Differentiation: Derivatives of Inverse Trig Functions

- This method enables us to find the derivatives of the inverse trigonometric functions arcsin, arccos and arctan.
- Recall the function $\arcsin\colon [-1,1] \longrightarrow \mathbb{R}$ satisfies

$$\sin(\arcsin(x)) = x$$

for $x \in [-1,1]$ and has graph:



5.4 Implicit Differentiation: Derivatives of arcsin

- To find the derivative of $y = \arcsin(x)$, we rewrite this relation as
$$\sin(y) = x \qquad \text{for } x \in [-1, 1]$$

- Implicit differentiation with respect to x gives:

- And we can write $\cos(y)$ in terms of x:

5.4 Implicit Differentiation: Derivatives of arcsin

- Substituting this into the implicit derivative gives:

Theorem 5.32

$$\arcsin'(x) = \frac{1}{\sqrt{1-x^2}}\ , \qquad \textit{for}\ \ x \in (-1,1).$$

Homework 48

How can we be sure that the identity $\cos(y) = \sqrt{1-\sin^2(y)}$ *holds in the previous calculation?*

5.4 Implicit Differentiation: Derivatives of arcsin

Homework 49

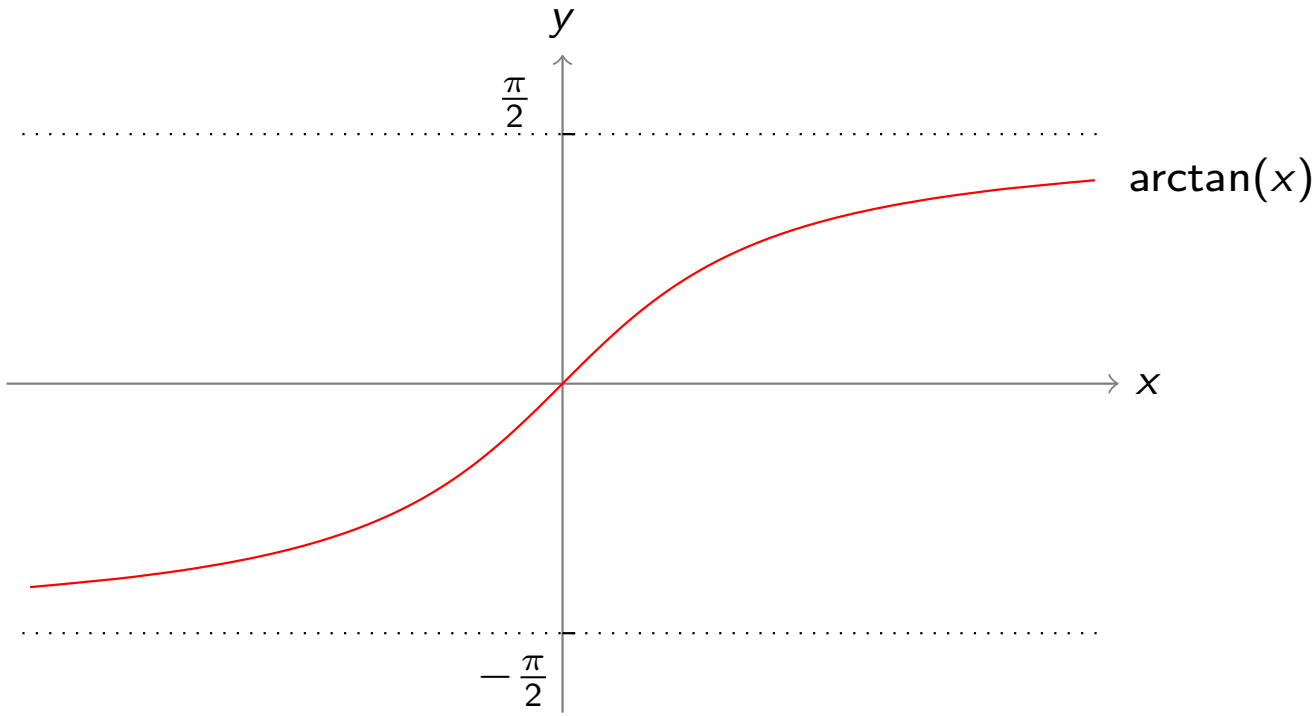
What can we say about the tangent line and hence derivative of arcsin *when* $x = -1$ *and* $x = 1$*?*

Homework 50

Find $\dfrac{d}{dx}\big(\arcsin(2x+1)\big)$.

5.4 Implicit Differentiation: Derivatives of arctan

Recall the function $\arctan\colon \mathbb{R} \longrightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with graph



We now use the identity $\sec^2(x) = \tan^2(x) + 1$ to find its derivative.

5.4 Implicit Differentiation: Inverse Trig Derivatives

Example 5.33

1. Check that $\sec^2(x) = \tan^2(x) + 1$ for all $x \in \mathrm{dom}(\tan)$.
2. Use (1) and implicit differentiation to find $\arctan'(x)$.

5.4 Implicit Differentiation: Inverse Trig Derivatives

Theorem 5.34

$$\arctan'(x) = \frac{1}{1+x^2}.$$

5.4 Implicit Differentiation: Inverse Trig Derivatives

Homework 51

Find $\frac{d}{dx}\left(\arctan(x^5)\right)$.

5.4 Implicit Differentiation: Derivative of arccos

Homework 52

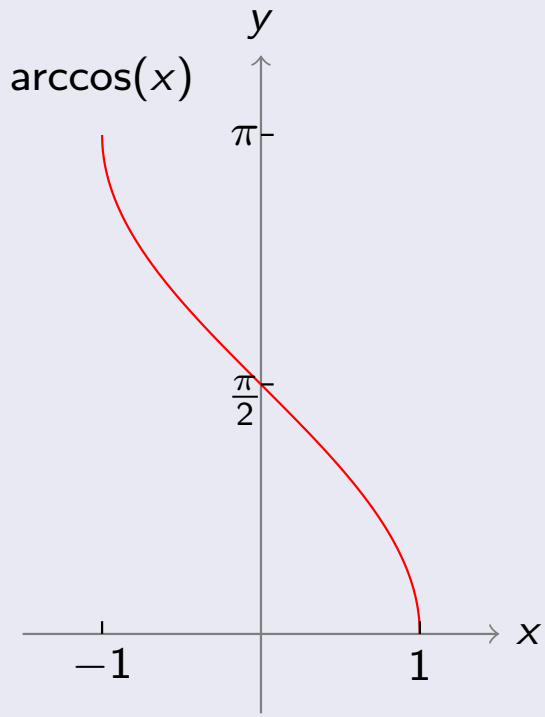
Consider the function

$$\arccos\colon [-1,1] \longrightarrow [0,\pi]$$

with graph as shown. Use implicit differentiation to show that

$$\frac{d}{dx}\big(\arccos(x)\big) = \frac{-1}{\sqrt{1-x^2}} = -\arcsin'(x)$$

for $-1 < x < 1$.
What happens when $x = -1$ *and* $x = 1$*?*



5.4 Implicit Differentiation: Derivative of arccos

Homework 53

Find $\dfrac{d}{dx}\big(\arccos(e^x)\big)$.

5.5 Differentiating Parametric Curves

5.5 Differentiating Parametric Curves

- The familiar definition of the derivative of real-valued function extends naturally to a parametric function.

Definition 5.35

Let $\mathbf{r} : I \longrightarrow \mathbb{R}^2$ where I is an interval. The derivative of $\mathbf{r}$ at t is defined by the limit:

$$\mathbf{r}'(t) = \lim_{h \longrightarrow 0} \frac{1}{h}\big(\mathbf{r}(t+h) - \mathbf{r}(t)\big).$$

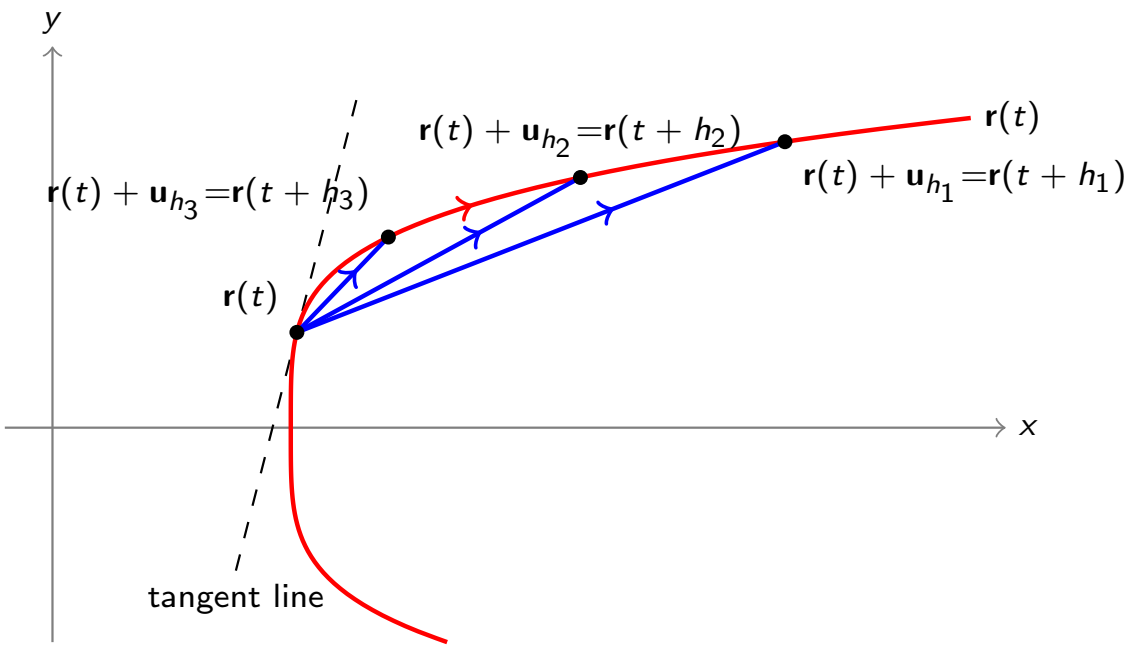
If this limit exists, $\mathbf{r}$ is said to be differentiable at t.

- Since $\mathbf{v}_h = \frac{1}{h}(\mathbf{r}(t+h) - \mathbf{r}(t))$ is a vector and $\mathbf{r}'(t)$ is a limit of the $\mathbf{v}_h$'s, it is a vector:

$$\mathbf{r}'(t) = a\mathbf{i} + b\mathbf{j}.$$

5.5 Differentiating Parametric Curves: Tangency

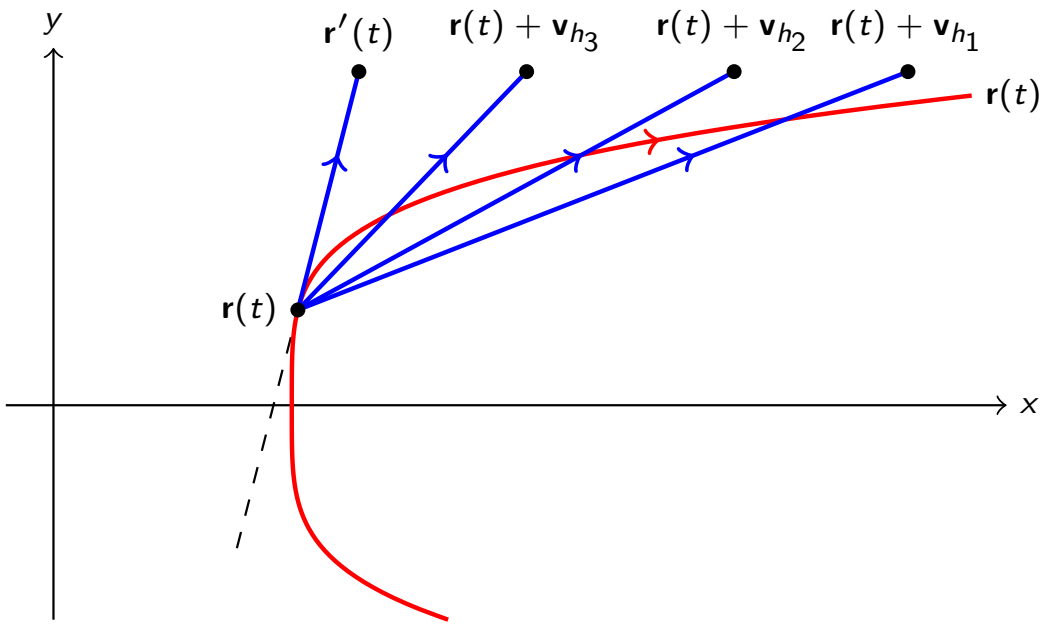
- Basing the vector $\mathbf{u}_h = \overrightarrow{AB}$ where $A = \mathbf{r}(t)$ and $B = \mathbf{r}(t+h)$ at $\mathbf{r}(t)$ gives a chord to the curve:



- As $h \longrightarrow 0$, the direction of $\mathbf{u}_h$ approaches the direction of the tangent line at $\mathbf{r}(t)$.

5.5 Differentiating Parametric Curves: Tangency

- Moving $\mathbf{v}_h = \frac{1}{h}(\mathbf{r}(t+h) - \mathbf{r}(t))$ to $\mathbf{r}(t)$ does not generally give a chord because scaling by $\frac{1}{h}$ changes the length ...

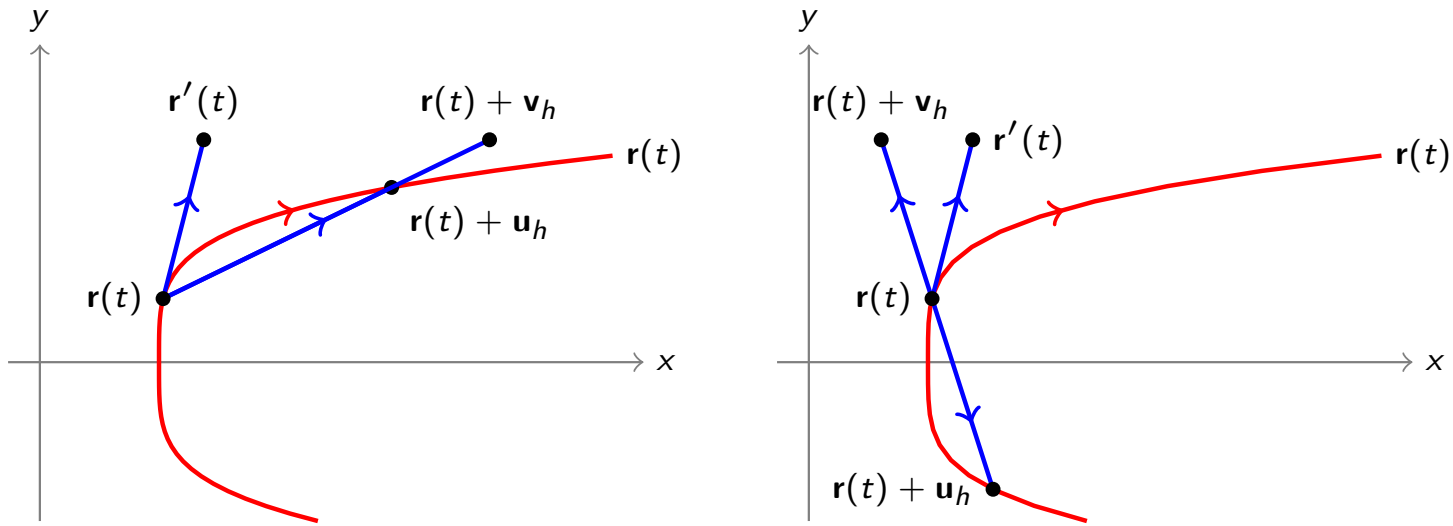


... but the direction of $\mathbf{v}_h$ still approaches the tangent.

- The vector $\mathbf{r}'(t)$ is still tangent to the curve at $\mathbf{r}(t)$.

5.5 Differentiating Parametric Curves

- $\mathbf{u}_h = \mathbf{r}(t+h) - \mathbf{r}(t)$ and $\mathbf{v}_h = \frac{1}{h}(\mathbf{r}(t+h) - \mathbf{r}(t))$ are parallel. For $h > 0$, they "follow" the curve's direction of motion.



- When $h < 0$, the vector $\mathbf{u}_h$ points opposite to the curve's direction of motion ...
- ... but $\mathbf{v}_h = \frac{1}{h}\mathbf{u}_h$ and $\frac{1}{h} < 0$, so $\mathbf{v}_h$ follows the direction of motion again!

5.5 Differentiating Parametric Curves

Theorem 5.36

Let $\mathbf{r} : I \longrightarrow \mathbb{R}^2$ *with* $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$. *Provided* x *and* y *are differentiable*

$$\mathbf{r}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j}.$$

- This makes differentiation very easy because we already know how to differentiate the functions $x : I \longrightarrow \mathbb{R}$ and $y : I \longrightarrow \mathbb{R}$.

Example 5.37

For each curve, give a formula for $\mathbf{r}'(t)$:

1. $\mathbf{r}(t) = (t^2 + 1)\mathbf{i} + (\frac{1}{3}t^3 + 1)\mathbf{j}$.
2. $\mathbf{r}(t) = \mathbf{i} + 3t^3\mathbf{j}$.
3. $\mathbf{r}(t) = 2t\mathbf{i} - (3t^3 + 1)\mathbf{j}$.
4. $\mathbf{r}(t) = 2\cos(t)\mathbf{i} + \sin(t)\mathbf{j}$.

5.5 Differentiating Parametric Curves

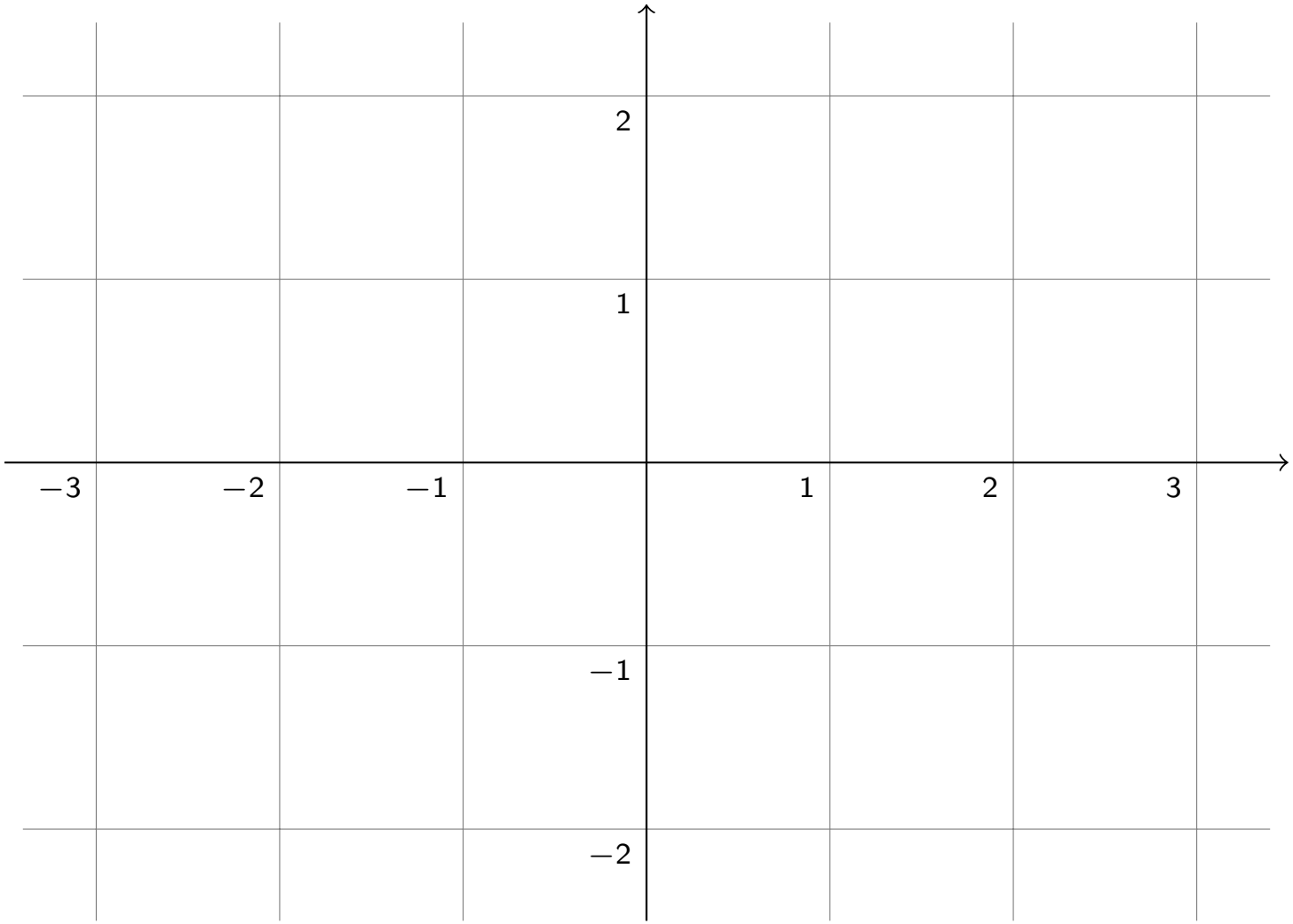
Example 5.38

For the curve given by $\mathbf{r}(t) = 2\cos(t)\mathbf{i} + \sin(t)\mathbf{j}$, calculate the vectors:

(a) $\mathbf{r}'(0)$ (b) $\mathbf{r}'\left(\frac{\pi}{2}\right)$ (c) $\mathbf{r}'(\pi)$ (d) $\mathbf{r}'\left(\frac{3\pi}{2}\right)$

and sketch them based at the appropriate point on the curve.

5.5 Differentiating Parametric Curves



Homework 54

For the parametric curve $\mathbf{r}(t) = (t^4 + 1)\mathbf{i} + t\mathbf{j}$, *calculate the vectors*

(a) $\mathbf{r}'(-1)$ *(b)* $\mathbf{r}'(0)$ *(c)* $\mathbf{r}'\left(\frac{1}{2}\right)$ *(d)* $\mathbf{r}'(1)$

and sketch them based at the appropriate point on the curve.

5.5 Differentiating Parametric Curves: Velocity and Speed

- For a particle whose motion is described by a curve $\mathbf{r}$, the vector $\mathbf{r}'(t)$ gives the <u>instantaneous velocity</u> at time t.
- This vector quantity expresses the velocity in the $\mathbf{i}$ and $\mathbf{j}$ directions at time t.
- The length $\|\mathbf{r}'(t)\|$ of the velocity vector gives the speed of motion along the curve:

$$\begin{aligned}\text{Speed} &= \|\mathbf{r}'(t)\| = \|x'(t)\mathbf{i} + y'(t)\mathbf{j}\| \\ &= \sqrt{(x'(t))^2 + (y'(t))^2}\end{aligned}$$

- Speed measures distance covered per unit time along the curve.
- This corresponds to our intuitive idea of the speed at which an object is moving at a given instant in time.

5.5 Differentiating Parametric Curves: Velocity and Speed

Example 5.39

For each curve, give a formula for the speed at time t:

1. $\mathbf{r}(t) = (t^2 + 1)\mathbf{i} + (\frac{1}{3}t^3 + 1)\mathbf{j}$
2. $\mathbf{r}(t) = 2t\mathbf{i} - (3t^3 + 1)\mathbf{j}$
3. $\mathbf{r}(t) = 2\cos(t)\mathbf{i} + \sin(t)\mathbf{j}$

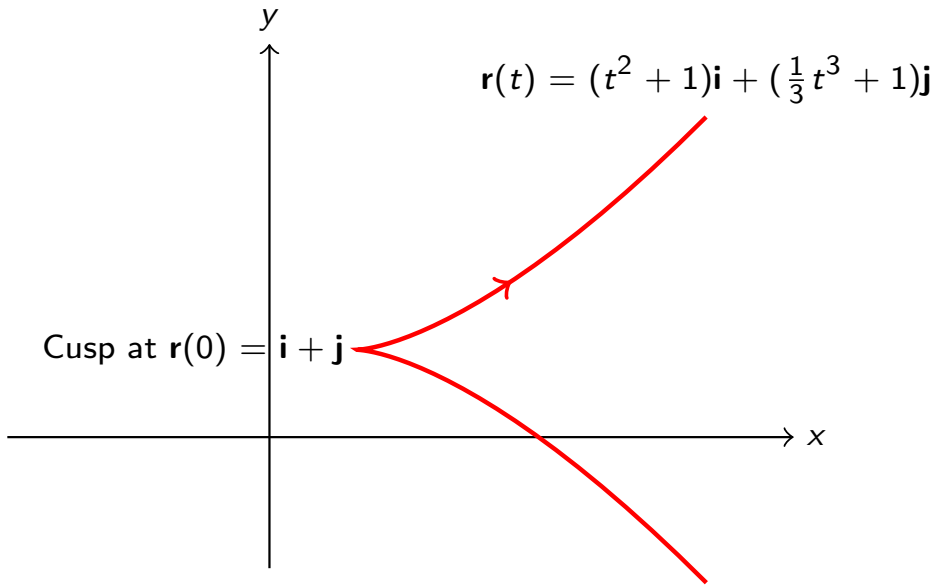
and find any times at which the speed is 0.

Example continued

5.5 Differentiating Parametric Curves: Cusps

- One of the curves in Example 5.39 had a point where the velocity was $\mathbf{0}$ (and hence the speed was 0).

- **Q:** What is the tangent line in this situation?
- **A:** There isn't one!



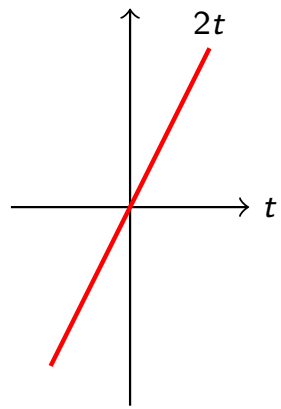
- If $\mathbf{r}'(t) = \mathbf{0}$ for some t, the curve usually has a sharp point called a <u>cusp</u> at $\mathbf{r}(t)$.
- Despite this, $\mathbf{r}$ is differentiable at 0.

5.5 Differentiating Parametric Curves: Cusps

The curve $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ may have a cusp at t when $x'(t) = y'(t) = 0$ and at least one of x' and y' change sign at t. We will not discuss the precise definition of cusps and exactly when a cusp exists. Below we examine a couple of curves where $\mathbf{r}'(t) = \mathbf{0}$ for some t.

Example 5.40 (optional)

Find all possible cusps for the curve $\mathbf{r}(t) = (t^2 + 1)\mathbf{i} + (\frac{1}{3}t^3 + 1)\mathbf{j}$.



Example 5.41 (optional)

Does the curve given by $\mathbf{r}(t) = t^3\mathbf{i} + t^5\mathbf{j}$ have a cusp?

5.5 Differentiating Parametric Curves: Cusps

Homework 55 (optional)

Find possible cusps of the curve given by

$$\mathbf{r}(t) = (t - \sin(t))\mathbf{i} + (1 - \cos(t))\mathbf{j}.$$

5.5 Differentiating Parametric Curves: Acceleration

- Since $\mathbf{r}' : I \longrightarrow \mathbb{R}^2$, we can differentiate again to obtain $\mathbf{r}'' : I \longrightarrow \mathbb{R}^2$.

Definition 5.43

At each t the vector $\mathbf{r}''(t)$ is called the <u>acceleration vector</u>.

- The vector $\mathbf{r}''(t)$ expresses the rate of change in velocity in the $\mathbf{i}$ and $\mathbf{j}$ directions at time t.

5.5 Differentiating Parametric Curves: Acceleration

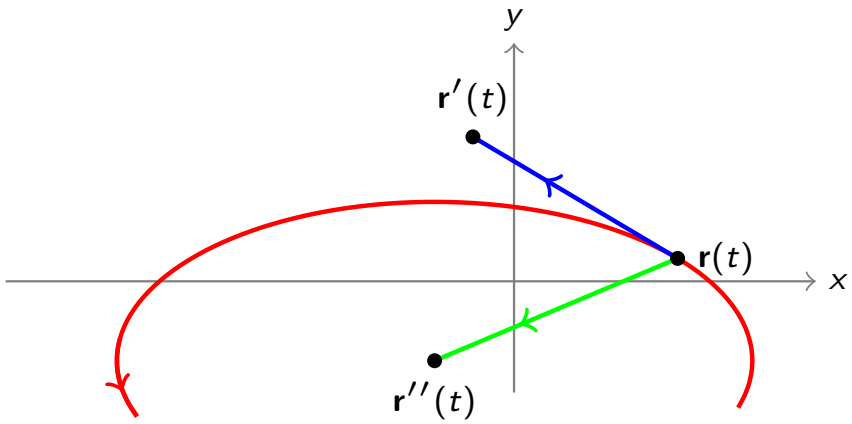
Example 5.44

For each curve, give a formula for the acceleration at time t:

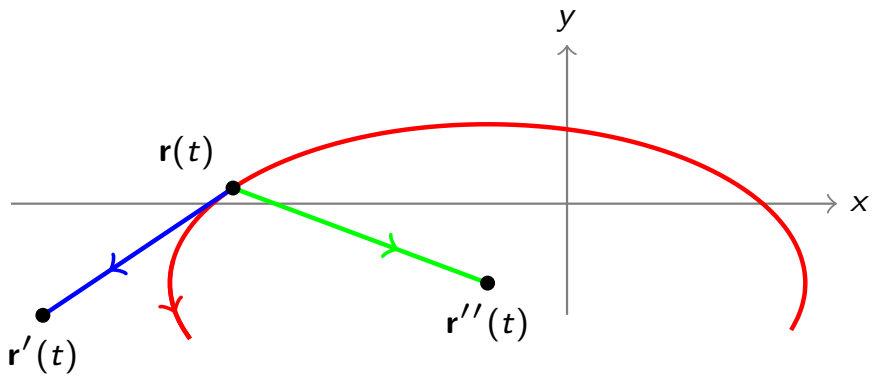
1. $\mathbf{r}(t) = (t^2 + 1)\mathbf{i} + (\frac{1}{3}t^3 + 1)\mathbf{j}$,
2. $\mathbf{r}(t) = 2t\mathbf{i} - (3t^3 + 1)\mathbf{j}$,
3. $\mathbf{r}(t) = (t - \sin(t))\mathbf{i} + (1 - \cos(t))\mathbf{j}$.

5.5 Differentiating Parametric Curves: Acceleration and Velocity

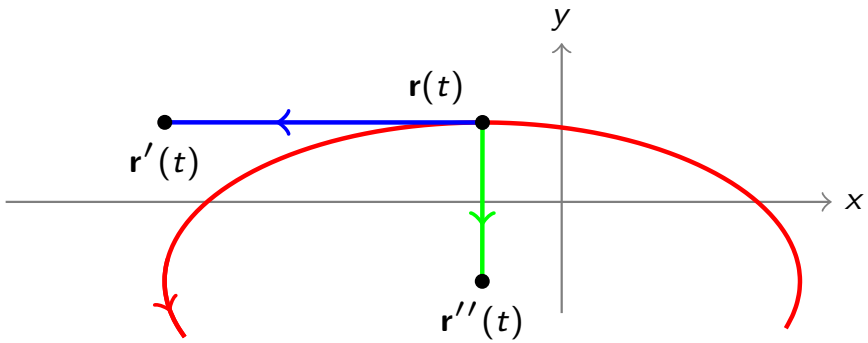
- When the angle between $\mathbf{r}''(t)$ and $\mathbf{r}'(t)$ is acute, the speed is increasing.



- When this angle is obtuse, the speed is decreasing.



- When $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$ are perpendicular, the speed is at a turning point.



5.5 Differentiating Parametric Curves: Acceleration and Velocity

- These results about speed are consequences of the following Theorem.

Theorem 5.45

Suppose $\mathbf{r}'(t) \neq \mathbf{0}$. *The speed function* $s(t) = \|\mathbf{r}'(t)\|$ *is:*

- *Decreasing if* $\mathbf{r}'(t) \cdot \mathbf{r}''(t) < 0$.
- *At a stationary point if* $\mathbf{r}'(t) \cdot \mathbf{r}''(t) = 0$.
- *Increasing if* $\mathbf{r}'(t) \cdot \mathbf{r}''(t) > 0$.

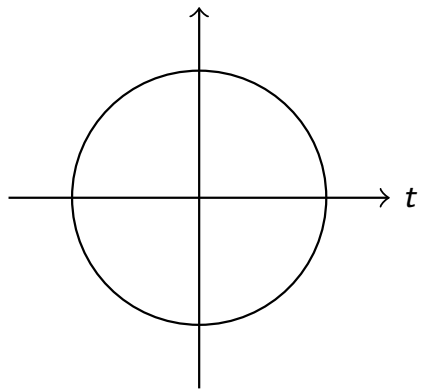
- This theorem is proved by determining the sign of

$$0 = \frac{d}{dt}\left[\|\mathbf{r}'(t)\|\right] = \frac{d}{dt}\left[\sqrt{(x'(t))^2 + (y'(t))^2}\right]$$

Example 5.46

For each curve, find the t values for which the speed is increasing:

1. $\mathbf{r}(t) = (t^2 + 1)\mathbf{i} + (\frac{1}{3}t^3 + 1)\mathbf{j}$
2. $\mathbf{r}(t) = (t - \sin(t))\mathbf{i} + (1 - \cos(t))\mathbf{j}$.



5.5 Differentiating Parametric Curves: Acceleration and Velocity

Homework 56

For a particle with motion given by $\mathbf{r}(t) = \cos(t)\mathbf{i} + 3\sin(t)\mathbf{j}$ *find the times at which:*

1. *The speed is at a maximum.*
2. *The speed is at a minimum.*

Express your answers as sets.